Microwave Spectroscopy

* State the criteria for a molecule to be microwave active. Give examples of at least four molecules which may show rotational microwave absorption spectrum. F-2020 [2(a)], F-2012, F-2015 (2)

Ans: Criteria for a molecule to be microwave active:

1. ~~Molecules must possess a permanent dipole moment~~

1. Molecules with permament dipole moments are ~~microwave~~ microwave active e.g. heteronuclear diatomic molecules.

2. Dipole moment should be changed during the rotation of the molecule.

Examples:
HCl, H_2O, CH_3Cl, CH_2Cl_2, CO etc.

* Which of the following molecules will have a microwave rotational spectrum?

F-2021 (2a), F-2018,

H_2, HCl, CH_4, CH_3Cl, CH_2Cl_2, H_2O, SF_6 (2)

Ans: Molecules ~~which~~ with permanent dipole moment are ~~radio~~ microwave active e.g. asymmetric (heteronuclear) molecule.

HCl, CH_3Cl, CH_2Cl_2, H_2O have a net non zero dipole moment. So, they are microwave active;

H_2, CH_4 and SF_6 have net ~~no~~ zero dipole moment. So, they are not microwave active;

* Classify the following molecules on the basis of the principle of moment of inertia:

HCl, CO_2, SiH_4, SF_6, NH_3, CH_3Cl, H_2O, CH_3OH.

F-2021, F-2018

HCl, CH_4, CH_3F, H_2O, OCS, BCl_3 in F-2016

HCl → Linear

CO_2 → Linear

SiH_4 → Spherical top

SF_6 → Spherical top

NH_3 → Symmetric top

CH_3Cl → Symmetric top

H_2O → Asymmetric top

CH_3OH → Symmetric top

CH_4 → Spherical top

CH_3F → Symmetric top

OCS → Linear

BCl_3 → Symmetric top

* Mention the limitations of microwave spectroscopy. F-2020, F-2017, F-2015,

Ans: Limitations of microwave spectroscopy for structure determination:

1. Only molecules with permanent dipole moments can be studied.

2. Large molecules have so many intermolecular distances that these quantities can not be sorted out from three determined moments of inertia.

3. Molecules should be in the gaseous form where the rotation is free and obeys the expression -

$$\varepsilon_J = \frac{h}{8\pi^2 I c} J(J+1)\ cm^{-1}$$

4. Generally gaseous sample is preferable. But for solid and liquid high pressure is needed.

5. Only bond angle and bond length can be determined by microwave spectroscopy.

* Write down the energy expression in cm-1 for a diatomic molecule based on rigid rotator model. Show that the absorption spectrum of a rigid rotator is expected to consist of a series of equidistant lines. F-2021, F-2018, F-2016

Ans:

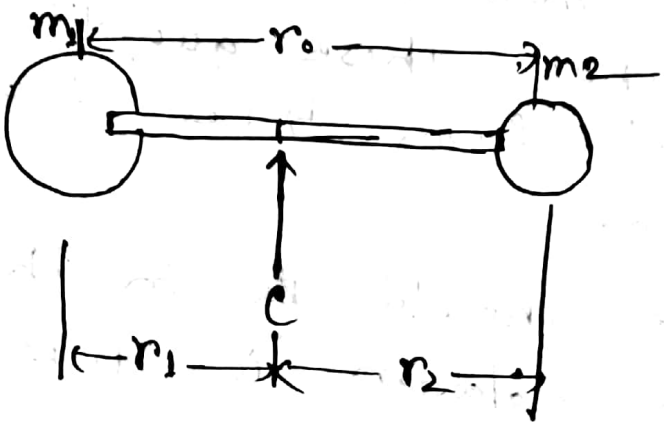


Fig: Rigid diatomic molecule treated as two masses m_1 and m_2

Consider a rigid diatomic molecules with masses m_1 and m_2 are joined ~~together~~ by a rigid bare whose length is,

$$r_0 = r_1 + r_2 \quad \text{---------(1)}$$

The molecules rotate end-over-end about a point C; the centere of gravity and hence the balancing equation is

$$m_1 r_1 = m_2 r_2 \quad \text{——— (2)}$$

The moment of inertia about C is given by,

$$I = m_1 r_1^2 + m_2 r_2^2$$

$$= m_2 r_2 r_1 + m_1 r_1 r_2$$

$$\therefore I = r_1 r_2 (m_1 + m_2) \quad \text{——— (3)}$$

But from (1) and (2);

$$m_1 r_1 = m_2 r_2 = m_2 (r_0 - r_1)$$

$$\therefore r_1 = \frac{m_2 (r_0 - r_1)}{m_1} \quad \frac{m_2 r_0}{m_1 + m_2}$$

or similarly, $r_2 = \frac{m_1 r_0}{m_1 + m_2}$

Replacing the values of r_1 and r_2 into eqn(3) we get,

$$I = \frac{m_1 m_2 r_0^2}{(m_1 + m_2)^2} \times (m_1 + m_2)$$

$$\Rightarrow I = \frac{m_1 m_2}{m_1 + m_2} \times r_0^2$$

$$\therefore I = \mu r_0^2$$

Here, μ = The reduced mass of the system

By the use of Schrodinger equation it may be shown that the rotational energy levels allowed to diatomic molecules are given by:

$$E_J = \frac{h^2}{8\pi^2 I} J(J+1) \text{ joules where } J = 0, 1, 2, 3 \text{ ---} \quad \text{---(a)}$$

where, J = The rotational quantum number

h = Planck's constant

I = Moment of inertia

Now, converting eqn (a) into wave number unit —

$$\epsilon_J = \frac{E_J}{hc}$$

$$= \frac{h}{8\pi^2 I c} J(J+1) \text{ cm}^{-1}, \quad J = 0, 1, 2 \text{ ---}$$

$$\therefore \boxed{\epsilon_J = BJ(J+1) \text{ cm}^{-1}} \quad \text{---(b)}$$

where,

$$B = \frac{h}{8\pi^2 I c} = \text{Rotational constant}$$

This is the energy expression for a diatomic molecule in cm^{-1} based on rigid rotator model.

Putting $J = 0, 1, 2, 3 \cdots$ etc in eqn (5) we get,

~~$\varepsilon_J = 0$~~

when $J = 0$, $\varepsilon_J = 0 \text{ cm}^{-1}$

when $J = 1$, $\varepsilon_J = 2B \text{ cm}^{-1}$

when $J = 2$, $\varepsilon_J = 6B \text{ cm}^{-1}$

According to selection rule, for rigid diatomic molecule $J = \pm 1$. It means that, transition can occur from a particular level to its intermediate neighbouring state.

Now,

for $J=0$ to $J=1$ transmition we get,

$$\varepsilon_{J=1} - \varepsilon_{J=0} = 2B - 0 = 2B \text{ cm}^{-1}$$

And, therefore,

$$\bar{\upsilon}_{J=0 \to J=1} = 2B \text{ cm}^{-1}$$

for, $J=1 \longrightarrow J=2$ transmition

$$\bar{\upsilon}_{J=1 \to J=2} = 6B - 2B = 4B \text{ cm}^{-1}$$

In general to raise the molecule from the state J to $(J+1)$ we get.

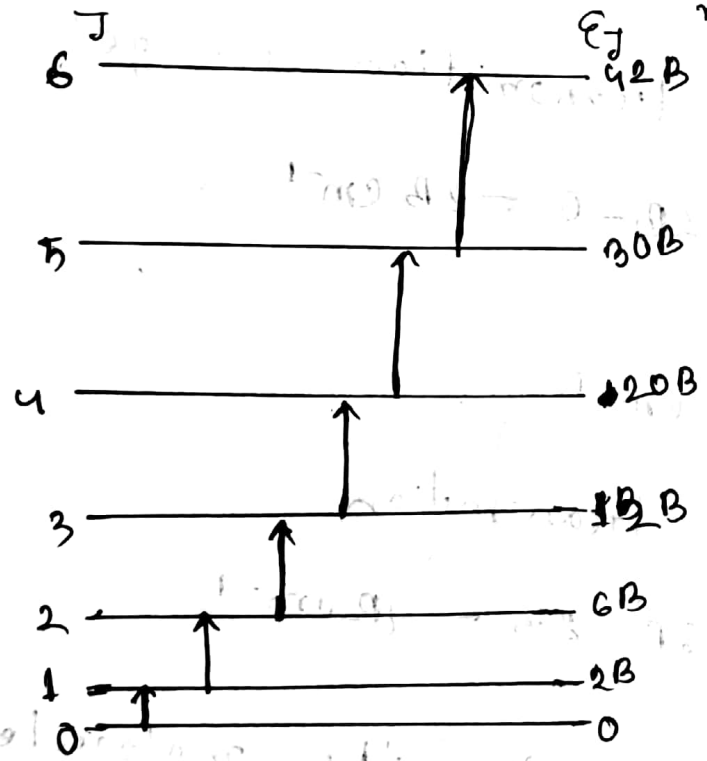
$$\bar{\nu}_{J \to J+1} = \varepsilon_{J+1} - \varepsilon_J$$

$$= hB(J+1)(J+2) - hBJ(J+1)$$

$$= hB(J+1)[J+2-J]$$

$$= 2B(J+1)\ cm^{-1}$$

Thus a stepwise raising of the rotational energy results in an absorption spectrum consisting of lines at 2B, 4B, 6B ... The distance between two consecutive rotational line is 2B which is constant. That means the absorption spectrum of a rigid rotator is expected to consist of a series of equidistant lines.



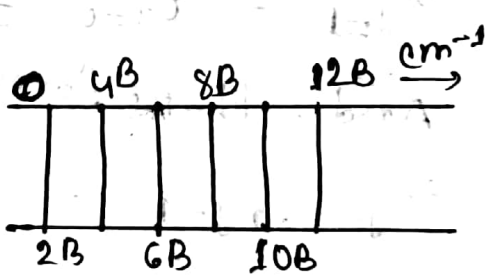


Fig-1: Allowed transmission between the energy levels of a rigid diatomic molecule and the spectrum which arises from them

* Write the energy expression and show allowed transitions between rotational energy levels for rigid diatomic molecule. Also show that absorption spectrum of rigid diatomic molecule consists of lines at 2B, 4B, 6B, 8B, ... cm^{-1}.

Ans: ~~Previous~~ আগেরটার মতো হবে।

* What are the limitations of the rigid rotator model? Write down the energy expression for non-rigid rotator in cm^{-1}. Compare the energy spacing between these two models. Discuss the isotope effect on rotational spectra.

F-2021, F-2018, F-2016

Ans:

Limitations of rigid rotator model:

1. The bond between two atoms is considered as rigid bond in rigid rotator model which ~~But~~ is not correct since all bonds are elastic to some extent.

2. It is considered that the separation between successive lines are $2B\ cm^{-1}$ which is

not correct. The separation between successive lines decreases steadily with increasing J.

For CH_3F,

when $J=0$ $B = 20.56\ cm^{-1}$

when $J=1$ $B = 20.48\ cm^{-1}$

when $J=2$ $B = 20.43\ cm^{-1}$

3. Here vibration is not considered though during one rotation there occures 1000 vibration.

Energy expression for non-rigid rotator:

Energy expression for non-rigid rotator in joules is:

$$E_J = \frac{h^2}{8\pi^2 I} J(J+1) - \frac{h^4}{32\pi^4 I^2} J^2(J+1)^2 \text{ joules}$$

And the expression in cm^{-1} is,

$$\epsilon_J = \frac{E_J}{hc} = BJ(J+1) - DJ^2(J+1)^2\ cm^{-1}$$

where, B = Rotational constant

D = Centrifugal distortion constant

Here,

$$D = \frac{h^3}{32\pi^4 I^2 r r k c} \text{ cm}^{-1}$$

for anharmonic force field, the expression becomes:

$$\varepsilon_J = BJ(J+1) - DJ^2(J+1)^2 + HJ^3(J+1)^3 + KJ^4(J+1)^4 \ldots \text{cm}^{-1}$$

where H, K etc, are small constant depend upon the geometry of molecule.

Energy spacing of non-rigid rotator and compare it with rigid rotator:

We know the energy expression for non-rigid rotator is;

$$\varepsilon_J = BJ(J+1) - DJ^2(J+1)^2 \text{ cm}^{-1}$$

Now, transition from J to J+1.

$$\varepsilon_{J+1} - \varepsilon_J = \bar{\upsilon}_{J \to J+1} = [B(J+1)(J+2) - BJ(J+1)] - [D(J+1)^2(J+2)^2 - DJ^2(J+1)^2]$$

$$= [B(J+1)\{J+2-J\}] - [D(J+1)^2\{(J+2)^2 - J^2\}]$$

$$= 2B(J+1) - 2D[(J+1)^2 - (J^2+2J+1-J)]$$

$$= 2B(J+1) - 4D(J+1)^3 \text{ cm}^{-1} \quad \text{---(2)}$$

So, the energy spacing between two successive lines are not exactly $2B\ cm^{-1}$, it is smaller than $2B\ cm^{-1}$.

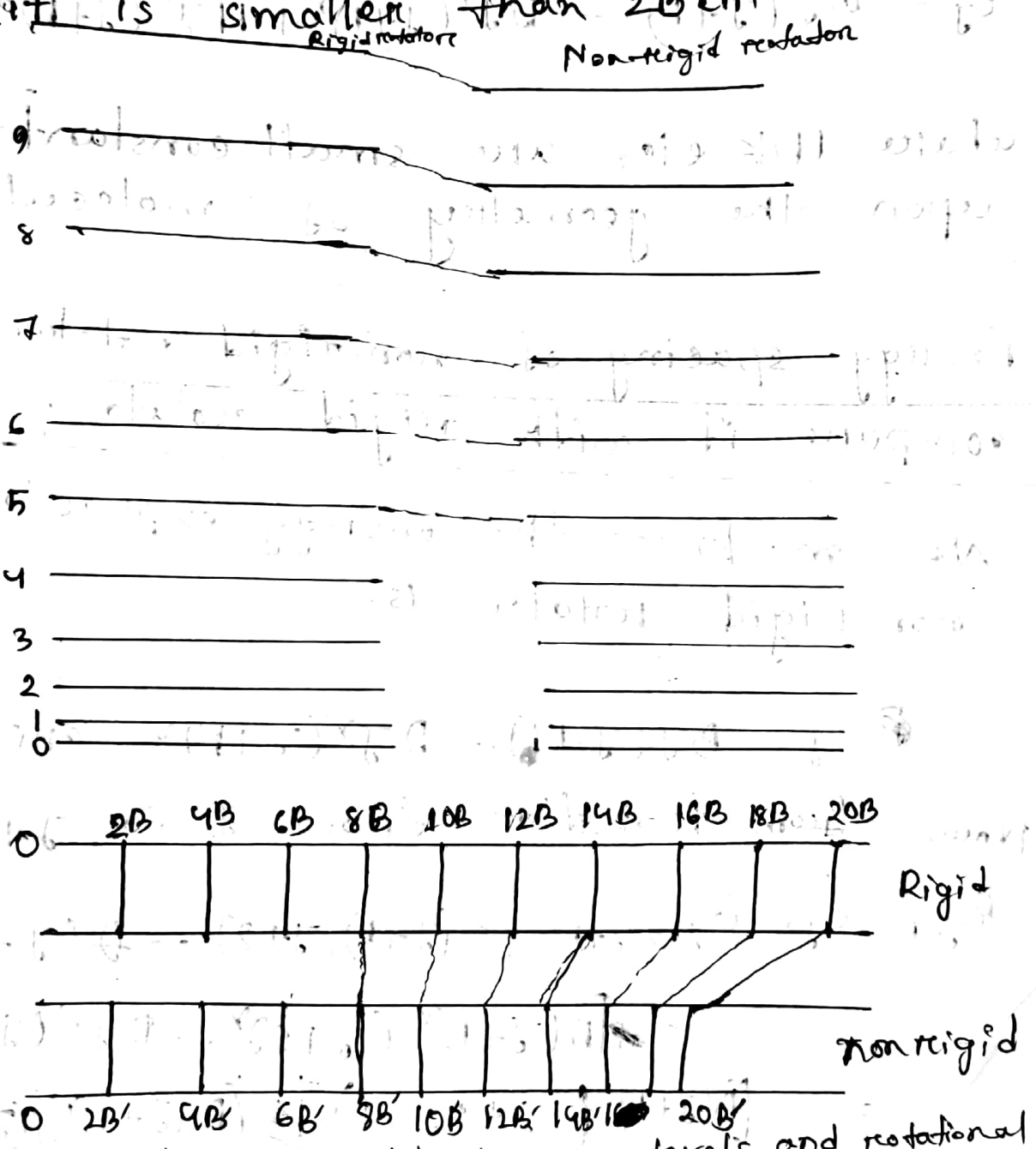


Fig: The change in rotational energy levels and rotational spectrum when passing from a rigid to non-rigid diatomic molecule.

The transition energy expression for rigid rotation is,

$$\epsilon_{J \to J+1} = 2B\,J\,(J+1)\ cm^{-1}$$

And the energy spacing is at $2B\ cm^{-1}$ distance between two successive energy level.

* Discuss the effect of isotope on rco

* Discuss the effect of isotope on rotational spectra. F-2021, 2018, ~~2016~~

* Explain the effect of isotopic substitution on rational spectra of a diatomic molecule such as carbon monoxide with the help of appropriate figures and equations.

F-2016

* Discuss the effect of isotopic substitution on the energy levels and spectrum of a diatomic molecule with example. FU-20??

Ans:

The effect of isotopic substitution:

When a particular atom in a molecule is replaced by its isotope a change in total mass and hence a change in moment of inertia and rotational constant B is obtained. Since the resulting substance is chemically identical with the original; there is no appreciable change in intermolecule distance.

Considering carbon monoxide as an ~~exa~~ example, we see that on going

from $^{12}C^{16}O$ to $^{13}C^{16}O$ atom there is a mass increase and hence a decrease in the B value. If we designate the rotational constant of $^{13}C^{16}O$ molecule with the symbol B' then B > B'. This change will reflected in the rotational energy level of the molecule and rotational spectrum which is shown in fig-1.

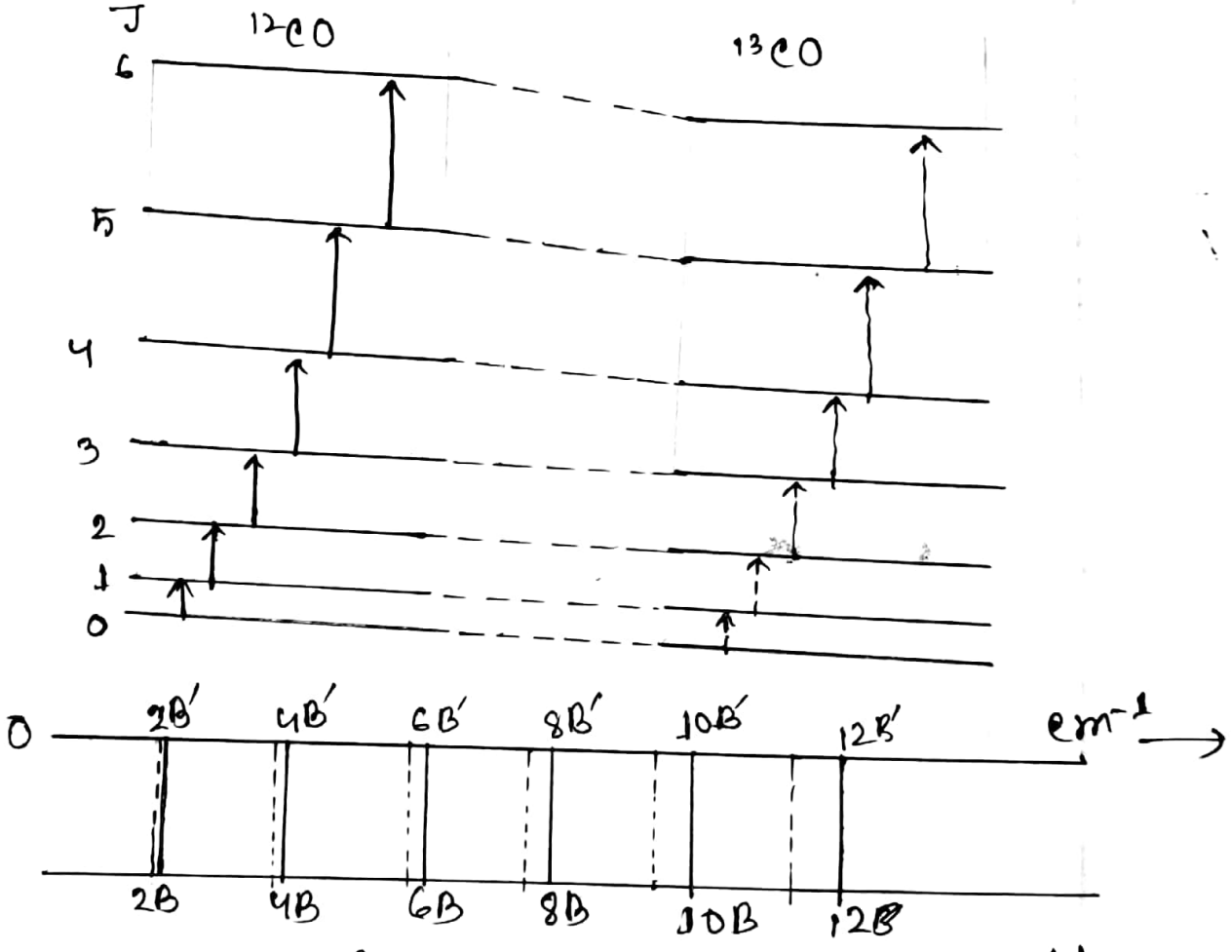


Fig: The effect of isotopic substitution on the energy levels and hence rotational spectrum of a diatomic molecule such as CO.